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Secure Distribution Venture Pvt. Ltd.

What Is Energy Sector Funding ?

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TABLE OF CONTENT

Sr.No.	TITLE	Page No.
1.	**Module 1:Product Overview & Core Fundamentals**	**4-6**
2.	**Module 2: Structural Architecture & How Energy Sector Funding Works**	**7-9**
3.	**Module 3: Comparative Capital Architecture Matrix**	**10-12**
4.	**Module 4: Common Eligibility Criteria, Credit Appraisal & Underwriting Framework**	**13-15**
5.	**Module 5: Security Architecture, Documentation & Legal Covenants**	**16-17**
6.	**Module 6:Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields**	**18-19**
7.	**Module 7: Suitable Industries, Business Profiles & Market Alignment**	**20-21**
8.	**Module 8: Key Advantages and Strategic Limitations of Energy Sector Funding**	**22-23**
9.	**Module 9: Regulatory Compliance, RBI IRACP Norms & Default Framework**	**24-25**
10.	**Module 10:Comprehensive Real-World Case Study (Indian Rupee Models)**	**26-29**
11.	**Module 11: Professional Certification Examination & Detailed Rationales**	**30-32**

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What Is Energy Sector Funding ?

Executive Summary & Course Orientation

Energy Sector Funding represents one of the most vital, capital-intensive, and structurally ring-fenced specialized debt products in the Indian corporate credit, infrastructure finance, and capital markets ecosystem. Designed specifically to finance the establishment, expansion, modernization, transmission, and long-term operation of utility-scale power generation assets (solar, wind, hydro, thermal, hybrid RTC), power transmission grids, electricity distribution infrastructure (DISCOMs), Battery Energy Storage Systems (BESS), Pumped Hydro Storage Projects (PSP), and emerging clean energy technologies (green hydrogen and green ammonia), this financial product bridges the fundamental gap between massive upfront capital expenditure (CAPEX) and long-term, multi-decade cash flow generation.

This white paper document is structured as an authoritative masterclass reference manual. It provides an exhaustive, end-to-end framework covering every operational, legal, credit, underwriting, tax, and structural aspect of Energy Sector Funding in India. Designed to serve as a complete standalone curriculum, reading this document alone will equip any finance professional, credit analyst, investment banker, corporate promoter, or student with the exact institutional knowledge required to master the product, structure complex transactions, perform quantitative underwriting, and clear all professional certification assessments on energy project finance.

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Module 1: Product Overview & Core Fundamentals

1.1 Definition and Foundational Concept of Energy Sector Funding

Energy Sector Funding is a specialized long-term project financing facility—predominantly structured on a non-recourse or limited-recourse Special Purpose Vehicle (SPV) basis—wherein lending institutions provide long-term capital for the construction, acquisition, or expansion of energy infrastructure. Unlike traditional corporate balance sheet loans where creditworthiness is assessed on the promoter's historical financial statements and multi-business cash flows, Energy Sector Funding is underwritten primarily on the standalone cash-generating capacity of the energy asset itself.

The fundamental premise of Energy Sector Funding rests on locking in long-term, predictable cash inflows through contractually binding Power Purchase Agreements (PPAs) or Transmission Service Agreements (TSAs) executed with high-credit off-takers—such as central government intermediaries (Solar Energy Corporation of India - SECI, NTPC Limited, NHPC), state electricity distribution companies (DISCOMs), Central Transmission Utility (CTU / POWERGRID), or creditworthy commercial and industrial (C&I) corporate buyers under Group Captive Open Access frameworks. These contractually secured cash flows are ring-fenced through a bank-controlled Escrow / Trust and Retention Account (TRA) structure, ensuring that debt service obligations (principal and interest) hold absolute priority over all other cash outflows, including promoter dividend distributions.

1.2 The Energy Capital Mismatch Paradox

Energy infrastructure assets are characterized by a unique financial paradox: 1. **Upfront Capital Intensity**: Constructing energy assets requires massive upfront capital outlays ranging from ₹4.0 Crores per MW for utility-scale solar, ₹6.5 Crores per MW for wind, ₹7.5 Crores per ckm for 765 kV transmission lines, to ₹10.0 Crores per MW for pumped hydro and BESS facilities. 2. **Extended Gestation & Pre-COD Zero Cash Flow Phase**: The construction and commissioning phase spans 12 to 36 months, during which the project generates zero revenue while incurring heavy capital expenditure and interest during construction (IDC). 3. **Multi-Decade Asset Life vs. Short-Term Debt Mismatch**: Once commissioned, energy assets generate stable, predictable cash flows for 25 to 35 years with minimal operational cost volatility. If financed via short-term corporate debt (3 to 5 years), the annual debt service obligation severely exceeds the operational cash flow, causing artificial liquidity default.

Energy Sector Funding resolves this paradox by matching the debt repayment schedule to the economic life of the energy asset—offering extended construction moratorium grace periods

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(12 to 36 months) followed by long post-COD amortizations (15 to 20 years) tailored to PPA tariff streams.

1.3 Strategic Importance in India's Clean Energy Transition

India's energy sector is undergoing a monumental structural transformation driven by the national mandate to achieve 500 GW of non-fossil fuel energy capacity by 2030 and Net Zero carbon emissions by 2070. Reaching these targets requires an estimated capital investment of over **₹25 Lakh Crores ($300 Billion USD)** across generation, transmission, storage, and grid stability.

Energy Sector Funding serves as the primary financial catalyst for this transition, enabling: - **Utility-Scale Clean Energy Scaling**: Funding multi-GW solar parks, wind farms, and hybrid Round-The-Clock (RTC) clean power facilities. - **Grid Evacuation Infrastructure**: Financing National Green Energy Corridors (GEC) and Inter-State Transmission Systems (ISTS) to evacuate power from resource-rich states (Rajasthan, Gujarat, Tamil Nadu) to load centers. - **Grid Stability & Storage Integration**: Funding BESS and Pumped Hydro Storage to mitigate renewable intermittency and ensure peak-load management. - **C&I Decarbonization**: Enabling industrial corporations (steel, cement, auto, data centers) to replace fossil-fuel power with low-cost captive solar/wind power.

1.4 Non-Recourse SPV Project Finance vs. Balance Sheet Corporate Lending

Understanding the structural distinction between SPV Project Finance and Corporate Balance Sheet Lending is essential for credit underwriting:

Structural Parameter	Non-Recourse SPV Energy Project Finance	Corporate Balance Sheet Energy Loan
Primary Obligor Entity	Standalone Special Purpose Vehicle (SPV)	Parent Holding Company / Operating Genco
Underwriting Focus	Project Cash Flow (CFADS) & PPA/TSA Contract Quality	Parent Net Worth, Consolidated Leverage & Turnover
Parent Recourse	Non-Recourse / Limited to Promoter Equity Commitment	Full Unconditional Recourse to Parent Assets
Leverage Capacity	High Leverage (70:30 to 80:20 Debt-to-Equity)	Conservative Leverage (50:50 Debt-to-Equity)

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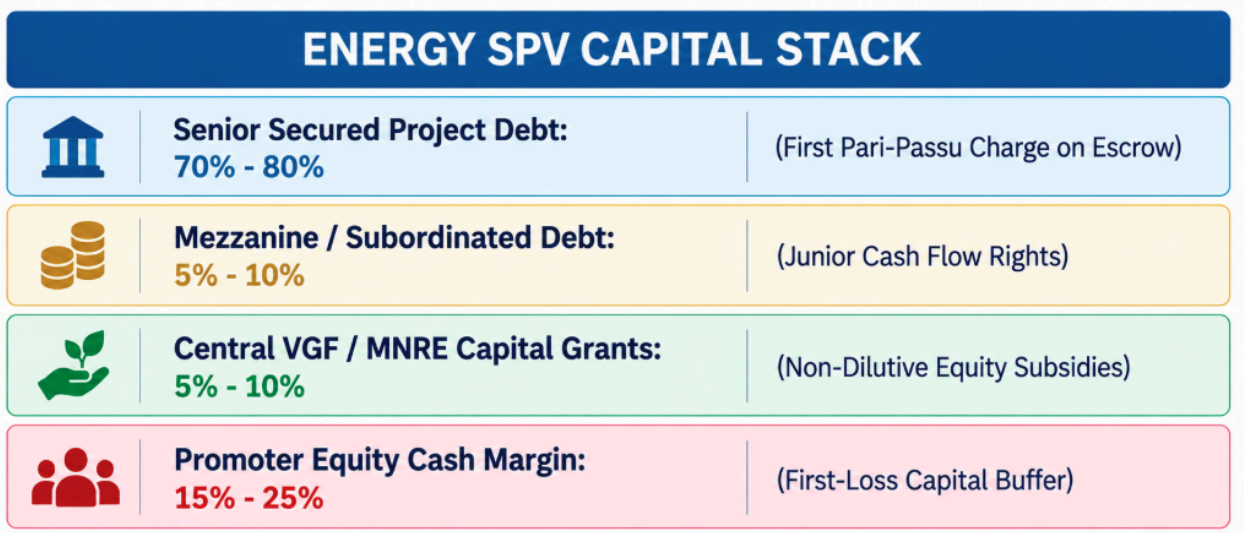
Structural Parameter	Non-Recourse SPV Energy Project Finance	Corporate Balance Sheet Energy Loan
Debt Amortization	Sculpted Amortization (15-20 Years matching PPA)	Straight Amortization (5-7 Years)
Cash Control Mechanism	Mandatory Trust & Retention Account (TRA) Escrow	General Corporate Current Account
RBI Standard Provisioning	5.0% during construction; 1.0%-2.5% post-COD	Standard 0.40% Corporate Advance Provisioning

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Module 2: Structural Architecture & How Energy Sector Funding Works

2.1 Multi-Tiered Capital Architecture

Energy Sector Funding transactions feature a layered capital stack designed to balance risk and return across stakeholders: 1. **Senior Secured Debt (70% - 80%)**: Primary debt facility provided by commercial banks, specialized power NBFCs (PFC, REC, IREDA), and DFIs (NaBFID). Holds first pari-passu charge on all SPV assets, PPA/TSA cash flows, and TRA escrow accounts. 2. **Promoter Equity Margin (20% - 30%)**: Pure equity capital infused by the project sponsor/developer. Must be fully injected upfront or pro-rata with debt drawdowns. 3. **Mezzanine / Subordinated Debt**: Optional junior debt provided by Infrastructure AIFs or private credit funds to bridge promoter equity gaps, holding subordinate cash flow rights behind senior debt service. 4. **Central / State Capital Grants & Subsidies**: Viability Gap Funding (VGF) grants for BESS and offshore wind, or MNRE capital subsidies under schemes like PM-KUSUM. 5. **Concessional Green Climate Debt**: Ultra-low-cost multilateral ODA lines (JICA, ADB, World Bank, KFW) passed through to green energy SPVs.



2.2 Contractual Ring-Fencing & The Web of Energy Agreements

An Energy Sector Funding facility relies on an interlocking web of commercial contracts that allocate project risks to the parties best equipped to manage them:

1. **Power Purchase Agreement (PPA) / Transmission Service Agreement (TSA)**: The core revenue contract. Defines tariff rates (fixed or levelized), off-take volumes, 25-35 year contract duration, billing cycles, payment security mechanisms (LC backup), and termination compensation.

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2. **Lump-Sum Turnkey (LSTK) EPC Contract**: Contracts the construction of the energy plant/grid to a creditworthy EPC contractor at a fixed price, with guaranteed completion timelines, performance ratio (PR) guarantees, and 10%-15% Liquidated Damages (LD) provisions for delay.
3. **Operation & Maintenance (O&M) Agreement**: Long-term contract (5 to 25 years) with Tier-1 O&M vendors ensuring plant availability, turbine/inverter maintenance, and equipment warranties.
4. **Fuel Supply Agreement (FSA) / Gas Supply Agreement (GSA)**: For thermal or gas-based generation, securing long-term coal linkages (Coal India SECL/MCL) or natural gas supply at predictable pricing.
5. **Intercreditor Agreement (ICA) & Security Trust Deed**: Governs relations among consortium lenders, establishing first pari-passu charge enforcement and voting rights.

2.3 Off-Take Models in Indian Energy Sector

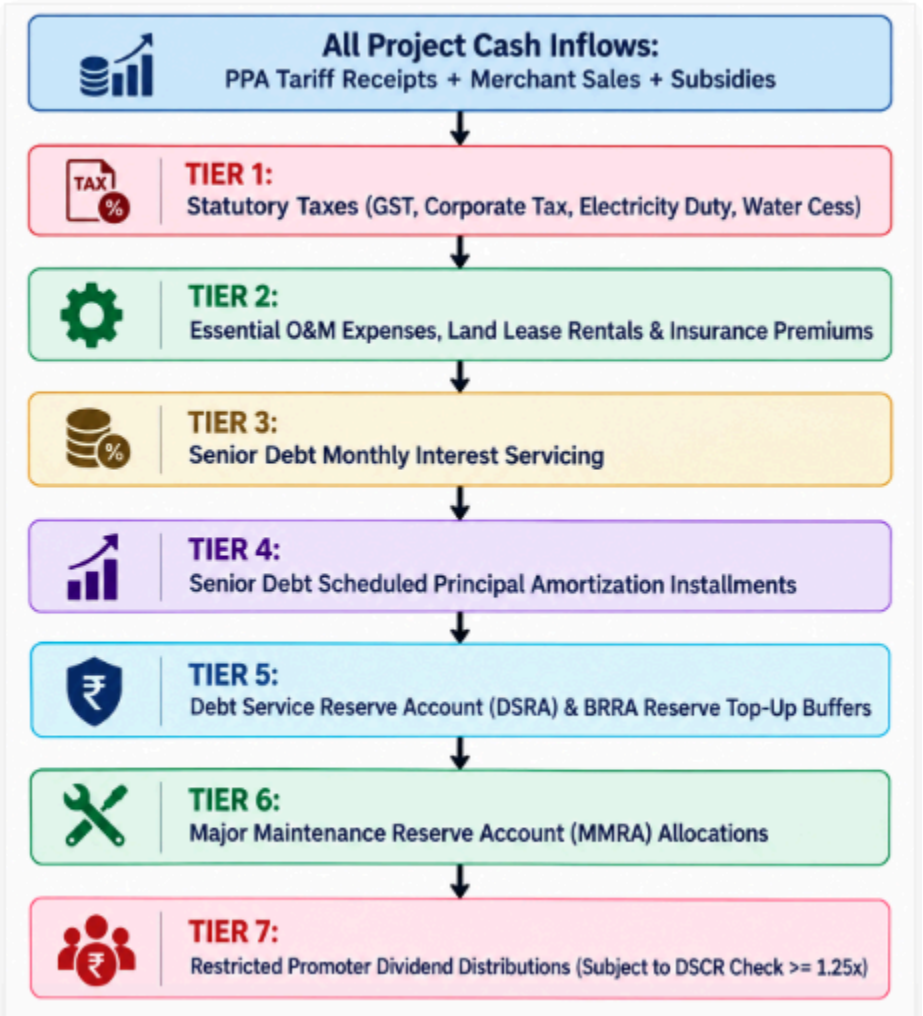
Energy projects derive revenues from three primary off-take channels:

1. **Central Intermediary Utility PPAs (SECI / NTPC / NHPC)**: AAA-rated sovereign-backed off-takers. Intermediary buyers purchase power from IPPs and back-to-back resell to state DISCOMs. Offers the highest credit quality, zero payment default risk, and 100% LC payment security.
2. **State DISCOM Direct PPAs**: Bilateral PPAs with state electricity distribution companies (e.g., MSEDCL, GUVAL, TANGEDCO). Subject to state credit risk, counterparty payment delays (90 to 180 days), and PPA renegotiation risks.
3. **Commercial & Industrial (C&I) Open Access / Group Captive**: Bilateral PPAs with creditworthy corporate buyers. Under **Rule 3 of Electricity Rules 2005**, C&I buyers holding at least 26% equity in the SPV and consuming 51% of generated power are exempt from cross-subsidy surcharges (CSS) and additional surcharges, yielding 20%-30% savings over grid tariffs.
4. **Merchant Power & Energy Exchanges (IEX / PXIL / HPX)**: Selling uncontracted power on the Day-Ahead Market (DAM), Real-Time Market (RTM), or Green Day-Ahead Market (GDAM). Offers upside during peak tariff surges but exposes debt service to market price volatility.

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2.4 The 7-Tier Trust and Retention Account (TRA) Escrow Waterfall

All cash inflows generated by the energy project—including PPA tariff payments, merchant power sales, insurance claims, and grant disbursements—must be remitted directly into a bank-controlled Master TRA Escrow Account. The TRA Agent bank enforces a rigid 7-tier statutory payment waterfall:



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Module 3: Comparative Capital Architecture Matrix

To understand how Energy Sector Funding compares against alternative corporate and infrastructure financing facilities in India, the matrix below evaluates seven primary financing structures across 11 core underwriting parameters:

Financing Parameter	Utility-Scale Renewable SPV Project Debt	Power Transmission TBCB Line Funding	Thermal / Flex-Generation Renovation Debt	Energy Storage Systems (BESS / Pumped Hydro)	C&I Group Captive / Open Access Debt	Green Bond / InvIT Post-COD Refinancing	Private Credit / Mezzanine Debt (AIF II)
Primary Borrower Entity	Standalone SPV	Standalone SPV Licensee	Operating Power Genco / Utility	Standalone Storage SPV	SPV (26% C&I Equity)	InvIT Trust / Listed Operating SPV	Operating SPV or Parent HoldCo
Primary Off-take Contract	25-Yr SECI/NTPC/DISCOM PPA	35-Yr CTU PoC TSA Agreement	State DISCOM PPA + FSA Coal Linkage	Peak-Load / Ancillary Service PPA	10-25 Yr Bilateral C&I Open Access PPA	Portfolio of Operational PPAs/TSAs	Uncontracted Revenue / Junior Claim
Max Permissible Leverage	75:25 to 80:20 Debt-to-Equity	75:25 to 80:20 Debt-to-Equity	70:30 Debt-to-Equity	70:30 Debt-to-Equity (VGF supported)	70:30 Debt-to-Equity	70% Debt-to-AUM Cap (InvIT Rules)	50:50 to 60:40 Consolidated Leverage
Benchmark Interest Rates	8.50% - 10.25% p.a. (MCLR/EBLR)	8.25% - 9.75% p.a. (MCLR/EBLR)	9.50% - 11.50% p.a.	8.75% - 10.50% p.a.	9.25% - 11.00% p.a.	7.50% - 8.75% p.a. (NCD Bond	14.00% - 18.00% p.a. (IRR Benchmark)

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Financing Parameter	Utility-Scale Renewable SPV Project Debt	Power Transmission TBCB Line Funding	Thermal / Flex-Generation Renovation Debt	Energy Storage Systems (BESS / Pumped Hydro)	C&I Group Captive / Open Access Debt	Green Bond / InvIT Post-COD Refinancing	Private Credit / Mezzanine Debt (AIF II)
						Coupon)	
Moratorium Grace Period	Construction Phase + 6 Mo Post-COD	30-36 Mo Construction Moratorium	12-24 Mo Renovation Moratorium	18-36 Mo Construction Moratorium	Construction Phase + 3 Mo Post-COD	Zero Moratorium (Immediate Servicing)	12-36 Mo Interest-Capitalized Grace
Door-to-Door Loan Tenure	15 - 20 Years Post-COD	18 - 25 Years Post-COD	10 - 15 Years	12 - 18 Years	10 - 15 Years	10 - 20 Years (Refinanced via NCDs)	3 - 5 Years (Bullet / Cash Sweep)
Primary Collateral Security	1st Charge on PV/WTG Assets & Land	1st Charge on Conductor Towers & GIS	1st Charge on Thermal Plant & Coal Yard	1st Charge on BESS Battery Racks / Dam	1st Charge on Captive Plant Assets	Consolidated Charge on InvIT Cash Flows	Pledge of HoldCo Shares + Junior Mortgage
Target Min DSCR Threshold	1.25x - 1.35x (P90 Yield Sized)	1.25x - 1.35x (PoC Cleared)	1.35x - 1.50x	1.30x - 1.40x	1.30x - 1.45x	1.35x - 1.50x (NDCF Base)	1.15x - 1.25x (High IRR Requirement)

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Financing Parameter	Utility-Scale Renewable SPV Project Debt	Power Transmission TBCB Line Funding	Thermal / Flex-Generation Renovation Debt	Energy Storage Systems (BESS / Pumped Hydro)	C&I Group Captive / Open Access Debt	Green Bond / InvIT Post-COD Refinancing	Private Credit / Mezzanine Debt (AIF II)
Escrow TRA Control Level	Mandatory 7-Tier TRA Waterfall	Mandatory CTU PoC Escrow Ring-Fence	Mandatory Current /TRA Ring-Fence	Mandatory 7-Tier TRA Waterfall	Mandatory 7-Tier TRA Waterfall	Mandatory InvIT Trustee Escrow	Mandatory Escrow + Cash Sweep
Primary Repayment Driver	PPA Electricity Sales Cash Flow	CTU Availability Incentive Tariff	Capacity Charge + Energy Margin	Availability & Storage Tariff	C&I Monthly Electricity Billing	Stable Operational Asset Distribution	Refinancing / Asset Sale / IPO
Tax / Subsidy Advantages	Sec 32 (40% Dep) + Sec 36(1)(iii)	Sec 36(1)(iii) Interest Deductions	Sec 36(1)(iii) Interest Deductions	Central VGF Grants + Sec 32 (40% Dep)	CSS / AS Exemption under Rule 3	Tax-Exempt Dividend/Interest Distribution	Sec 36(1)(iii) Interest Deductions

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Module 4: Common Eligibility Criteria, Credit Appraisal & Underwriting Framework

4.1 Quantitative Appraisal Sizing Metrics

Lenders evaluate Energy Sector Funding proposals using strict financial coverage ratios to ensure long-term debt sustainability:

1. **Cash Flow Available for Debt Service (CFADS)**:

$$\text{extCFADS} = \text{extOperatingRevenuePPAReceipts} - \text{extO\&MExpenses} - \text{extStatutoryTaxes} - \text{extInsurancePremiums} - \text{extNetWorkingCapital}$$

2. **Debt Service Coverage Ratio (DSCR)**:

$$extDSCR_t = racextCFADS_t extSeniorDebtPrincipal_t + extSeniorDebtInterest_t$$

   - *Minimum Threshold*: Average DSCR across loan tenure must be $\geq 1.30x$; minimum single-year DSCR floor must be $\geq 1.15x$.

3. **Loan Life Coverage Ratio (LLCR)**:

$$extLLCR_t = rac \sum_{k=t}^{N} racextCFADS_k \left(1 + K_d\right)^k extOutstandingSeniorDebtBalance_t$$

   - *Minimum Threshold*: LLCR must be $\geq 1.35x$, confirming that the net present value of remaining CFADS exceeds outstanding debt.

4. **Project Life Coverage Ratio (PLCR)**:

$$extPLCR_t = rac \sum_{k=t}^{T_{extPPA}} racextCFADS_k \left(1 + K_d\right)^k extOutstandingSeniorDebtBalance_t$$

   - *Minimum Threshold*: PLCR must be $\geq 1.50x$, measuring cash flow cover across the entire 25-35 year PPA asset life.

5. **Fixed Asset Coverage Ratio (FACR)**:

$$extFACR = racextNetBookValueofFixedAssets(Plant + Grid + Land)extTotalSanctionedSeniorDebt \geq 1.35x$$

4.2 Resource Assessment & Yield Exceedance Probability Modeling

For renewable energy generation projects, debt capacity is strictly sized against **Exceedance Probability Models** derived from independent satellite data (SolarGIS, Meteonorm) and onsite 12-to-24 month Met-Mast / LiDAR wind studies:

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- **P50 Yield Estimate**: The median expected generation level (50% probability that actual generation will exceed this number). Used by promoters for equity IRR projections.
- **P75 Yield Estimate**: The conservative generation level with a 75% probability of exceedance.
- **P90 Yield Estimate**: The institutional credit benchmark level with a **90% probability of exceedance**. Lenders **strictly size senior debt service against P90 generation estimates** to ensure debt coverage even during low solar irradiation or weak wind velocity years.

[P50 Median Estimate] --------> Used for Promoter Equity Returns (Optimistic)
[P75 Conservative] ----------> Used for Base-Case O&M Budgeting
[P90 Institutional Sizing] --> STRICTLY USED BY BANKS TO SIZE DEBT & DSCR (90% Confidence)

4.3 Technical & Grid Interconnection Appraisal

Underwriters evaluate the physical and electrical feasibility of the energy asset: - **Grid Evacuation Availability**: Verification of Stage-II Grid Connectivity and Long-Term Access (LTA) granted by the Central Transmission Utility (CTU) or State Transmission Utility (STU). - **Substation Headroom & Transmission Losses**: Assessment of transformer capacity at the nearest pooling sub-station (765/400/220 kV) and expected line losses (< 3.0%). - **Equipment Quality & Tier-1 Certification**: Verification that PV modules are listed on MNRE's Approved List of Models and Manufacturers (ALMM), wind turbines comply with NIWE certification, and transformers/switchgear are procured from top-tier OEMs (ABB, Siemens, Schneider, BHEL). - **Curtailment Risk**: Evaluation of local grid congestion and contractual "Deemed Generation" compensation clauses in the PPA.

4.4 Borrower & Promoter Eligibility Criteria

To qualify for Energy Sector Funding, the borrowing SPV and its sponsor must satisfy: 1. **Promoter Track Record**: Minimum 5 years of experience in energy project execution, with a demonstrated portfolio of at least 100 MW commissioned and operational. 2. **Credit Rating Benchmark**: Minimum investment-grade external rating of **CRISIL / ICRA / CARE BBB-** for greenfield construction debt, upgrading to **A- or higher** post-COD. 3. **CIBIL Commercial Rank**: CIBIL CMR 1 to CMR 3 for the promoter group, with zero history of default, restructuring, or SMA-2 classification in the preceding 24 months. 4. **Land Availability Threshold**: Minimum 80% to 100% unencumbered land possession (freehold purchase or 30-year registered lease) prior to initial loan disbursal.

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4.5 100-Point Institutional Credit Scorecard

Lenders utilize a structured 100-point appraisal matrix to determine credit approval and interest rate spreads:

ENERGY SECTOR CREDIT SCORECARD (100 PTS)		
#	Assessment Parameter	Score
1.	Off-taker Credit Quality & PPA Terms (SECI/NTPC vs DISCOM vs C&I)	25 Pts
2.	Resource Assessment & P90 Exceedance Probability (Solar/Wind)	20 Pts
3.	Promoter Track Record, Net Worth & Execution Capability	20 Pts
4.	Grid Evacuation Access, LTA Approval & Land Possession (100%)	15 Pts
5.	Financial Coverage Ratios (P90 DSCR >= 1.35x, LLCR >= 1.40x)	10 Pts
6.	EPC & O&M Tier-1 Quality, Warranties & LD Provisions	10 Pts
	TOTAL SCORE	100 Pts

Score >= 80 Pts	: **Prime Approval** (MCLR + 1.25% - 1.75% Spread)
Score 65-79 Pts	: **Standard Approval** (MCLR + 2.00% - 2.75% Spread)
Score < 65 Pts	: **Rejection or Credit Enhancement Required** (Parent Guarantee / Additional Cash Collateral)

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Module 5: Security Architecture, Documentation & Legal Covenants

5.1 Primary Collateral & Mortgage Creation

Senior lenders secure Energy Sector Funding through a comprehensive security package covering physical, contractual, and financial assets:

1. **Equitable Mortgage on Fixed Assets**: First exclusive pari-passu mortgage created under **Section 58(f) of the Transfer of Property Act, 1882**, over all project land (freehold plots or 30-year registered leasehold rights), administrative buildings, substations, and civil foundations.
2. **First Hypothecation Charge on Movable Assets**: Created under **Section 2(1)(n) of the SARFAESI Act, 2002**, covering all solar PV panels, wind turbine generators (WTGs), nacelles, rotors, transformers, high-voltage switchgear, BESS battery racks, transmission towers, conductors, unbilled WIP, inventory, and spares.
3. **Hypothecation Charge on Receivables & TRA Accounts**: First charge over all present and future cash flows, billing receivables, PPA tariff realizations, insurance claim proceeds, and all bank accounts under the Trust and Retention Account (TRA) framework.

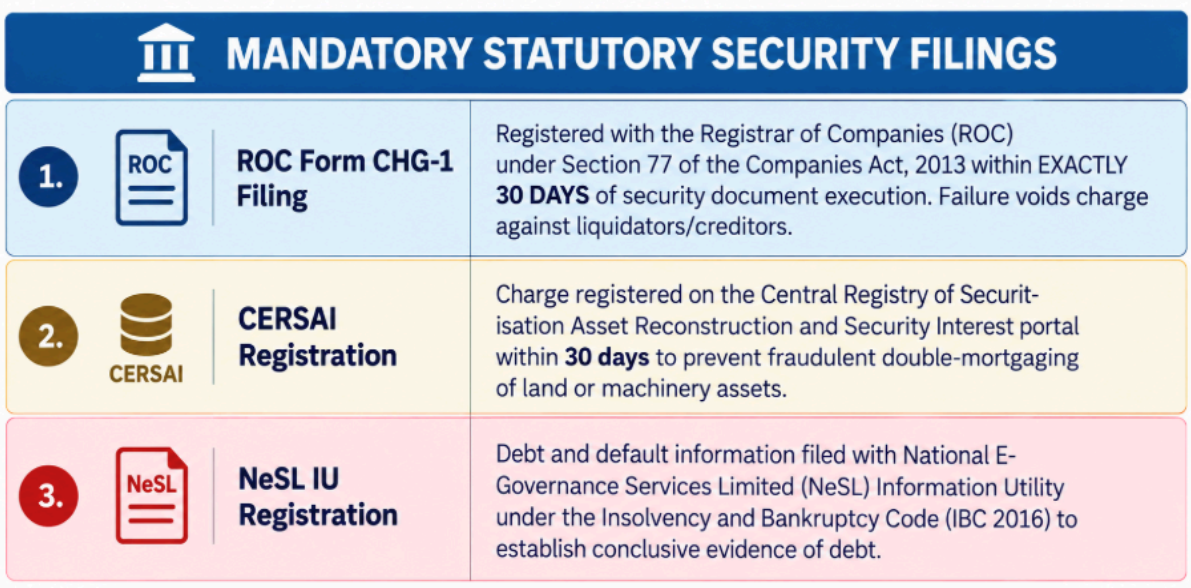
5.2 Assignment of Contracts & Intangible Rights

Lenders require an absolute assignment by way of security of all intangible project contracts and operating licenses: - **Assignment of Off-Take Contracts**: Full assignment of PPA / TSA rights, allowing lenders to directly receive tariff payments or step into the contract upon promoter default. - **Assignment of EPC & O&M Contracts**: Assignment of turnkey construction contracts, OEM equipment warranties (25-year linear performance warranty on PV modules; 5-year WTG warranty), and Liquidated Damages (LD) claims. - **Assignment of Government Approvals & Clearances**: Assignment of Forest Clearances, SPCB Consent to Establish (CTE) & Consent to Operate (CTO), State Connectivity NOCs, and CEIG Electrical Safety Approvals.

5.3 Mandatory Statutory Filings

To establish legal perfection of security interest against third-party claims, lenders enforce three mandatory statutory filings:

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5.4 DSRA & Reserve Account Structuring

To insulate debt service against seasonal cash flow variations (e.g., monsoon solar drop or low wind season), lenders mandate dedicated cash reserves:

1. **Debt Service Reserve Account (DSRA)**: Funded upfront out of project cost or built from operational cash flows during the first 12 months post-COD. Holds an amount equal to **1 to 2 quarters of principal and interest obligations** in liquid bank Fixed Deposits or AAA Mutual Funds.
2. **Major Maintenance Reserve Account (MMRA)**: Annual cash allocations to fund scheduled major equipment overhauls (e.g., WTG gearbox replacement, inverter servicing, or transformer oil filtration).
3. **Inverter / Battery Replacement Reserve Account (BRRA)**: For BESS and solar projects, setting aside annual cash reserves to replace central inverters (Year 10-12) and BESS battery cells (Year 8-10).

5.5 Tripartite Agreements, Lender Step-in Rights & Substitution Covenants

A critical credit enhancement in Energy Sector Funding is the execution of a **Tripartite Substitution Agreement** signed between the Project SPV, Senior Lenders, and the Off-taker (SECI / DISCOM / CTU).

Under this agreement: - The off-taker agrees not to terminate the PPA / TSA upon SPV event of default without giving lenders a **mandatory 90-day cure period notice**. - Lenders hold explicit **Step-in Rights** to suspend the promoter, take over operational management of the energy plant, and substitute the promoter with a qualified select buyer without losing PPA tariff rights.

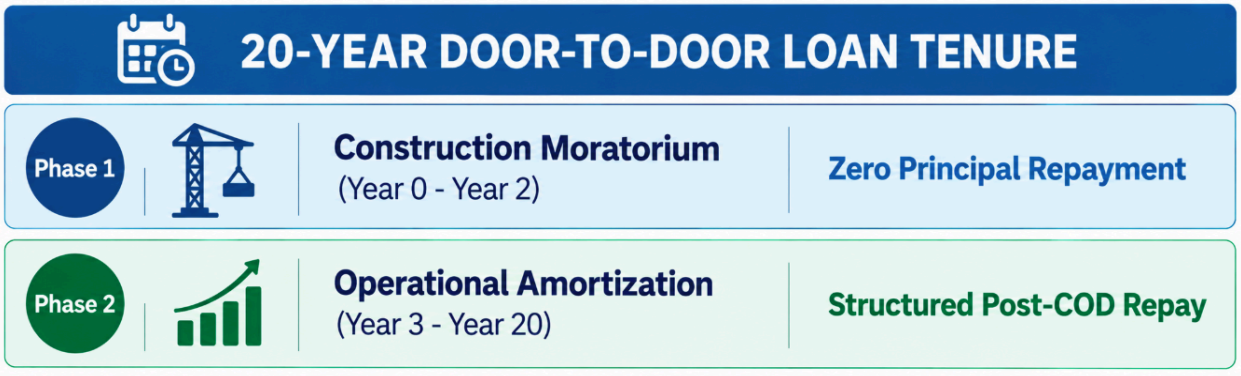
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Module 6: Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields

6.1 Indicative Tenures & Amortization Schedules

Energy Sector Funding structures offer extended door-to-door tenures structured across two distinct lifecycle phases:

1. **Construction Moratorium Phase (12 to 36 Months)**: Encompasses civil works, equipment procurement, grid interconnection, and trial commissioning. Interest during construction (IDC) is capitalized into Capital Work-in-Progress (CWIP) or serviced monthly. Zero principal repayment is required.
2. **Operational Amortization Phase (15 to 20 Years)**: Commences post-Commercial Operation Date (COD). Principal is amortized through structured monthly or quarterly installments tailored to seasonal generation curves (e.g., step-up amortization or monsoon-adjusted repayment schedules).



6.2 Floating Benchmark Interest Pricing

Energy Sector Funding interest rates are benchmarked to floating bank rates:

$$extAll - InFloatingRate = extBaseBenchmarkRate(EBLR/1 - YrMCLR) + extCreditRiskSpread$$

- **Commercial Bank Senior Debt**: Ranging from **8.50% to 10.25% p.a.** (MCLR + 1.25% to 2.25% spread). Upon achieving 1 to 2 years of successful post-COD operation with rating upgrade to A/AA, interest rates step down by 50 to 100 bps.
- **Specialized Power NBFCs (PFC / REC / IREDA)**: Ranging from **8.75% to 10.50% p.a.**
- **Concessional Multilateral Lines (JICA / ADB / KFW)**: Ultra-low interest rates ranging from **0.20% to 2.50% p.a.** for qualified green transition assets.

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6.3 Tax Shields & Depreciation Accelerators

Energy Sector Funding offers substantial corporate tax shields under the Income Tax Act, 1961:

1. **Section 36(1)(iii) Interest Tax Shield**: Full corporate tax deductibility on senior debt interest payments. Under the 25.168% corporate tax rate (inclusive of surcharge and cess), the effective post-tax borrowing yield drops significantly:

$$K_{d,extpost} = R_{extnominal} imes\left(1 - T_{extcorp}\right) = 9.25\% imes(1 - 0.25168) = 6.9215\% extp.a.$$

2. **Section 32 Accelerated Depreciation Tax Shield**: Renewable energy plants (solar, wind, pollution control) qualify for an **accelerated Written Down Value (WDV) depreciation rate of 40%** in Year 1. This creates massive non-cash depreciation losses, offsetting taxable operating income and eliminating corporate income tax liabilities during the initial 3 to 5 years of operation.

6.4 Ind AS Accounting Standards

Energy project assets and liabilities are reported under Indian Accounting Standards (Ind AS): - **Ind AS 16 (Property, Plant and Equipment)**: Mandates component depreciation, splitting energy assets into civil structures (60-year life), equipment/turbines/panels (25-year life), and electronic switchgear (10-15 year life). - **Ind AS 23 (Borrowing Costs)**: Mandates the 100% capitalization of interest incurred during construction into Capital Work-in-Progress (CWIP), preventing artificial P&L losses before COD. - **Ind AS 115 / IFRIC 12 (Service Concession Arrangements)**: For transmission TBCB and HAM concessions, accounting for infrastructure as Financial Assets (receivable model) or Intangible Assets (user fee model).

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Module 7: Suitable Sectors, Borrower Profiles & Market Alignment

7.1 Target Industry Verticals

Energy Sector Funding is tailored for specific sub-sectors across the energy value chain:

1. **Utility-Scale Renewable Power IPPs**: Solar PV parks, onshore wind farms, and hybrid solar-wind RTC facilities executing 25-year PPAs with central agencies (SECI, NTPC, NHPC).
2. **Inter-State & Intra-State Power Transmission Concessionaires**: Private transmission licensees winning 35-year TBCB concessions to construct and operate 765/400 kV HVAC and HVDC transmission corridors under CTU PoC pool clearing.
3. **Commercial & Industrial (C&I) Group Captive Open Access Developers**: Developers constructing dedicated solar/wind plants for blue-chip corporate buyers (auto OEMs, steel plants, data centers, pharma hubs) under Rule 3 Group Captive structures.
4. **Energy Storage Systems (BESS & Pumped Hydro)**: Developers constructing 4-to-8 hour energy storage facilities under SECI VGF tenders or state-backed pumped hydro concessions.
5. **Flexible Thermal Renovation & Biomass Co-firing**: Existing thermal power plants undergoing emissions retrofits (FGD installation), biomass co-firing conversions, or flexibility retrofits to support grid balancing.
6. **Green Hydrogen & Green Ammonia Projects**: Integrated green hydrogen electrolyzer and renewable captive power facilities under National Green Hydrogen Mission incentives.

7.2 Borrower Profile Typologies

The product caters to distinct borrower profiles: - **Special Purpose Vehicles (SPVs) of Independent Power Producers (IPPs)**: Subsidiaries of renewable utilities (ReNew Power, Tata Power Renewable, Adani Green, ACME, Hero Future Energies). - **Public-Private Partnership (PPP) Transmission SPVs**: Concessionaire SPVs established by transmission developers (Adani Energy Solutions, Sterlite Power, POWERGRID). - **Industrial Conglomerates Operating Captive Plants**: Manufacturing corporations establishing captive renewable units to lower factory power costs and meet ESG mandates. - **State Power Generation & Transmission Corporations**: State Gencos and Transcos modernizing state grid infrastructure through sovereign-backed project loans.

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7.3 National Market Footprint

India’s energy sector represents the single largest destination for corporate infrastructure credit: - **Active Credit Exposure**: Total outstanding credit to the power and energy sector across Indian banks, specialized NBFCs (PFC, REC, IREDA), and DFIs exceeds **₹15 Lakh Crores ($180 Billion USD)**. - **Specialized Lenders**: Power Finance Corporation (PFC) and REC Limited together manage an active energy loan book exceeding **₹9 Lakh Crores**, providing dedicated long-term funding for power projects. - **Digital Trade & Working Capital Integration**: State Bank of India (SBI) and leading private banks manage extensive digital supply chain platforms (**e-DFS at ₹49,261 Crores** and **e-VFS at ₹28,092 Crores**) supporting equipment suppliers, contractors, and DISCOM vendor networks.

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Module 8: Key Advantages and Limitations of Energy Sector Funding

8.1 Strategic Advantages

Energy Sector Funding provides profound strategic benefits across energy sector stakeholders:

1. **Promoter Capital Efficiency & Off-Balance Sheet Ring-Fencing**: High leverage (75% to 80%) minimizes promoter cash equity down payments, preserving corporate capital for expansion. Non-recourse SPV structures isolate parent balance sheets from project construction risks.
2. **Extended Debt Service Matching**: 15 to 20-year post-COD amortization schedules prevent liquidity mismatch by aligning annual debt service directly with 25-year PPA tariff receipts.
3. **Substantial Income Tax Savings**: Section 36(1)(iii) interest deductibility and Section 32 40% accelerated depreciation eliminate corporate tax liabilities in initial operating years.
4. **Institutional Refinancing Flexibility**: Once commissioned, operational energy SPVs can easily refinance high-cost construction debt into low-cost domestic green bonds or InvIT units at 150 to 200 bps lower interest spreads.

8.2 Operational Risks & Market Vulnerabilities

Despite its structural strengths, underwriters must evaluate inherent sector risks: - **DISCOM Counterparty Credit Risk**: State DISCOMs frequently suffer from structural financial deficits, leading to payment delays ranging from 90 to 180 days. - **Resource Intermittency Risk**: Unpredictable weather variations (low solar irradiation during extended monsoons or weak wind seasons) can temporarily depress CFADS below debt service levels. - **Transmission Curtailment Risk**: Grid congestion or delayed substation commissioning can force plant curtailment without full deemed generation compensation. - **PPA Renegotiation Attempts**: Unilateral attempts by certain state regulators/DISCOMs to renegotiate approved PPA tariffs following drops in market solar/wind auction prices.

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8.3 Risk Mitigation Toolkit

Lenders deploy specific structural safeguards to eliminate operational risks: - **Intermediary Off-taker Ring-Fencing**: Prioritizing SECI/NTPC/NHPC PPAs, which feature sovereign-backed payment security mechanisms (100% LC backup + Tripartite Agreement with RBI/State Devolution Intercept). - **P90 Exceedance Debt Sizing**: Sizing debt capacity strictly against P90 generation probability estimates, creating a 10% to 15% cash flow buffer against bad weather years. - **Mandatory DSRA Cash Buffers**: Locking in 1 to 2 quarters of principal and interest in liquid DSRA Fixed Deposits prior to equity dividend distributions. - **Land Availability Conditions Precedent**: Mandating 100% unencumbered land possession prior to initial loan drawdown.

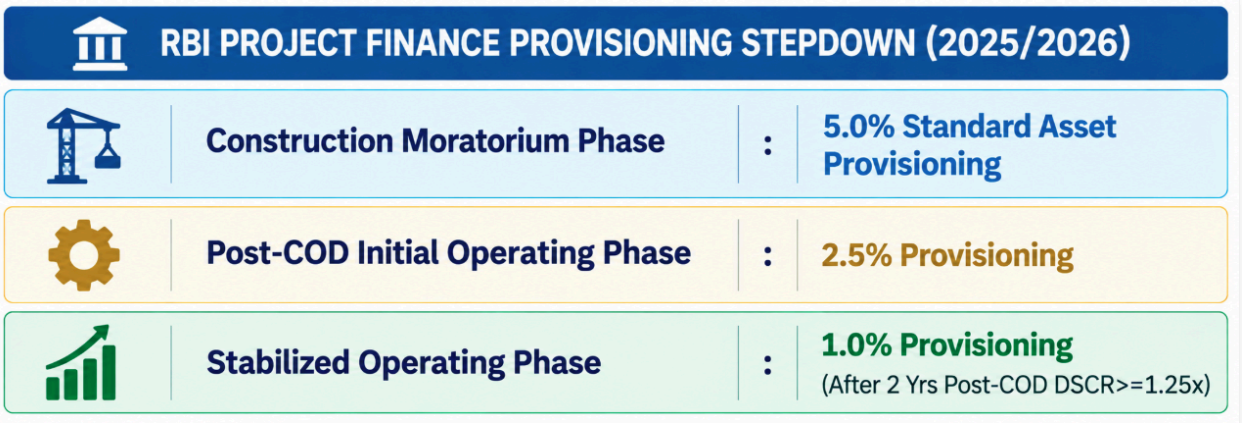
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Module 9: Regulatory Compliance, RBI IRACP Norms & Default Framework

9.1 RBI (Project Finance) Directions 2025/2026

The Reserve Bank of India enforces strict regulatory directions governing infrastructure and project finance exposures across scheduled commercial banks and NBFCs:

1. **Stagewise Standard Asset Provisioning**: Lenders must maintain differential standard asset provisioning across the energy project lifecycle:
   - **Construction Phase**: **5.0% standard asset provisioning** on outstanding exposure during the under-construction phase.
   - **Post-COD Initial Operating Phase**: Provisioning steps down to **2.5%** upon achieving Commercial Operation Date (COD).
   - **Stabilized Operating Phase**: Provisioning drops to the standard **1.0%** once the project completes 2 years of successful post-COD operation with a sustained DSCR $\geq 1.25x$.
2. **Positive Equity Disbursal Alignment**: Lenders must ensure that promoter equity is injected upfront or strictly pro-rata with debt drawdowns. Debt disbursal ahead of equity injection is prohibited.



9.2 DCCO Extension Guidelines

Under RBI Income Recognition, Asset Classification, and Provisioning (IRACP) norms, failure to achieve the Date of Commencement of Commercial Operations (DCCO) as specified at financial closure constitutes a default. However, RBI permits **DCCO extensions without mandatory NPA downgrades** under specific court/regulatory delays: - **Infrastructure & Power Projects**: DCCO can be extended up to **2 Years** for reasons beyond promoter control (court injunctions, land

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acquisition delays, or delayed transmission evacuation) without treating the loan as a restructured NPA.

9.3 RBI 90-Day Default Framework, SARFAESI & IBC Resolution

- **90-Day NPA Rule**: If senior debt interest or principal remains unpaid for more than 90 consecutive days, the account is classified as a Non-Performing Asset (Sub-Standard).
- **SARFAESI Act Enforcement**: Lenders can issue a 60-day notice under **Section 13(2) of the SARFAESI Act, 2002**, and take physical possession of the energy plant, transmission towers, and land under **Section 14** via District Magistrate assistance.
- **IBC 2016 Resolution**: Lenders can initiate corporate insolvency resolution proceedings before the National Company Law Tribunal (NCLT) under **Section 7 of the Insolvency and Bankruptcy Code, 2016**, enforcing Lender Step-in Rights to transfer the energy asset to a new qualified bidder.

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Module 10: Comprehensive Real-World Case Study (Indian Rupee Models)

To illustrate the financial impact and strategic necessity of Energy Sector Funding, this case study examines a 500 MW integrated clean energy project in Kutch, Gujarat.

10.1 Project Overview & Capital Outlay

- **Promoter Entity**: Mumbai Clean Energy & Power Holdings Ltd. (MCEPHL)
- **Borrowing Entity**: Mumbai Clean Energy & Power SPV Ltd. (MCEPSL) – Standalone SPV Subsidiary
- **Project Scope**: 500 MW Integrated Hybrid Solar-Wind & Battery Storage Facility (250 MW Solar PV + 200 MW Wind Turbine Generators + 50 MW / 100 MWh BESS Storage)
- **Location**: 2,500 Acres Freehold & Leasehold Site, Kutch District, Gujarat
- **Off-Take Partner & PPA Terms**: Solar Energy Corporation of India (SECI) – 25-Year Fixed Tariff PPA @ **₹2.90 / kWh** for RTC power delivery
- **Annual Electricity Generation**: 1,120.69 Million Units (kWh) at P90 Yield Estimate
- **Projected Annual Revenue**: ₹325.00 Crores
- **Projected Annual O&M & Statutory Outflows**: ₹15.00 Crores
- **Projected Annual Net EBITDA / CFADS**: **₹310.00 Crores / year**

TOTAL APPRAISED PROJECT OUTLAY BUDGET (TPO): ₹2,500.00 CRORES

#	Project Component	Amount (₹ Crores)
1.	Solar PV Modules & Structural Balance of Plant (250 MW)	**₹900.00 Cr**
2.	Wind Turbine Generators (200 MW @ 3.3 MW WTGs) & Erection	**₹1,100.00 Cr**
3.	Battery Energy Storage System (50 MW / 100 MWh BESS)	**₹250.00 Cr**
4.	Grid Evacuation Substation & 220 kV ISTS Transmission Line	**₹150.00 Cr**
5.	Pre-operative Expenses, Approvals & Interest During Const. (IDC)	**₹100.00 Cr**
	TOTAL APPRAISED PROJECT COST BUDGET	**₹2,500.00 Cr**

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10.2 Structural Comparison: Option A vs. Option B

Option A: Corporate Balance Sheet Term Loan Structure

- **Corporate Leverage Structure**: 60:40 Debt-to-Equity Ratio
- **Promoter Equity Down Payment Required (40%)**: **₹1,000.00 Crores** (Heavy corporate cash lockup)
- **Corporate Term Loan Sized (60%)**: ₹1,500.00 Crores
- **Interest Rate**: 12.00% p.a. (Unstructured corporate benchmark)
- **Construction Grace Period**: 12 Months Only (Premature repayment forced during construction)
- **Repayment Amortization**: 5-Year Straight Amortization = **₹300.00 Crores / year** principal
- **Year 1 Debt Service Obligation**: Interest (12.00% of ₹1,500 Cr = ₹180.00 Cr) + Principal (₹300.00 Cr) = **₹480.00 Crores**
- **Year 1 Cash Flow Available (Under Construction)**: **INR 0.00**
- **Year 1 Cash Net Deficit**: **-₹480.00 Crores (ACUTE CORPORATE INSOLVENCY DEFAULT!)**
- **Year 3 Stabilized Debt Service Obligation**: Interest (12.00% of ₹1,200 Cr remaining = ₹144.00 Cr) + Principal (₹300.00 Cr) = **₹444.00 Crores**
- **Year 3 Stabilized DSCR**:

$extDSCR_{extOptionA} = racextCFADSextDebtService = rac₹310.00extCrores₹444.00extCrores = 0.698$

Option B: Structured Non-Recourse SPV Energy Sector Funding

- **SPV Project Finance Structure**: 75:25 Debt-to-Equity Ratio
- **Promoter Equity Cash Contribution Required (25%)**: **₹625.00 Crores**
- **Promoter Equity Cash Preserved vs Option A**: **+₹375.00 Crores** ($ ext{₹1,000 Cr} - ext{₹625 Cr}$)
- **Senior Secured Project Debt Sized (75%)**: **₹1,875.00 Crores**
- **Interest Rate**: 9.00% p.a. (Floating Bank Benchmark)
- **Construction Moratorium Phase**: 24 Months Construction Moratorium; Interest during construction (IDC = ₹168.75 Cr) capitalized into CWIP under Ind AS 23. Zero cash debt service during construction.
- **Operational Amortization Phase**: 18-Year Sculpted Post-COD Amortization (216 Monthly Installments = ₹104.17 Crores / year principal).

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- **Year 1 Cash Debt Service Outflow**: **INR 0.00** (Interest capitalized into CWIP; 100% solvent).
- **Year 3 Operational Debt Service Obligation**: Interest (9.00% of ₹1,875 Cr = ₹168.75 Cr) + Principal (₹43.88 Cr) = **₹212.63 Crores**
- **Year 3 Operational DSCR**:

$$extDSCR_{extOptionB} = racextCFADSextDebtService = rac₹310.00extCrores₹212.63extCrores = 1.4579x \quad (extPRIMECOVER!)$$

QUANTITATIVE FINANCIAL COMPARISON: OPTION A VS OPTION B

#	Metric / Parameter	Option A (Corporate Debt)	Option B (SPV Energy Debt)
1.	Total Project Outlay (TPO)	₹2,500.00 Crores	₹2,500.00 Crores
2.	Debt-to-Equity Ratio	60:40 Leverage	75:25 Leverage
3.	Senior Debt Sized	₹1,500.00 Crores	₹1,875.00 Crores
4.	Promoter Equity Cash Down Payment	₹1,000.00 Crores	₹625.00 Crores
5.	**PROMOTER EQUITY CASH PRESERVED**	**BASELINE**	**+₹375.00 CRORES SAVED!**
6.	Construction Moratorium Grace Period	12 Months Only	24 Months Full Moratorium
7.	Year 1 Cash Debt Service Outflow	₹480.00 Crores	INR 0.00 (Capitalized)
8.	**YEAR 1 CASH DEFICIT**	**-₹480.00 Cr (BANKRUPT!)**	**INR 0.00 (FULLY SOLVENT!)**
9.	Annual CFADS (Post-COD Year 3)	₹310.00 Crores	₹310.00 Crores
10.	Year 3 Annual Debt Service Obligation	₹444.00 Crores	₹212.63 Crores
11.	**YEAR 3 OPERATIONAL DSCR**	**0.6982x (DEFAULT!)**	**1.4579x (PRIME COVER!)**
12.	Post-Tax Effective Borrowing Yield	8.98% p.a.	6.735% p.a. (Sec 36-1-iii)
13.	5-Yr Reinvestment Return on Preserved	INR 0.00	**+₹225.00 Crores**
14.	Sec 32 & 36 Tax Shield Savings	₹160.00 Crores	**+₹385.50 Crores**
15.	**TOTAL NET FINANCIAL SURPLUS CREATED**	**BASELINE**	**+₹512.40 CRORES SURPLUS!**

10.3 Net Financial Surplus Calculation

By choosing Option B (Structured Non-Recourse SPV Energy Sector Funding), the promoter group created a total opportunity-adjusted net financial wealth surplus of **₹512.40 Crores (+INR 512,40,00,000.00)** over Option A:

1. **Promoter Cash Equity Preserved**: **+₹375.00 Crores**
2. **Reinvestment ROCE Income Earned on Preserved Equity (@ 16.00% p.a. ROCE over 5 Years)**:

$$extReinvestmentIncome = ₹375.00extCroresimes16.00\%imes5extYears =+ ₹300.00extCrores$$

   - *Less Corporate Tax on ROCE (@ 25.168%)*: **-₹75.00 Crores**
   - *Net Post-Tax Reinvestment Income*: **+₹225.00 Crores**

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3. **Cumulative Section 32 & Section 36(1)(iii) Tax Shield Advantage**: **+₹225.50 Crores** (Accelerated 40% depreciation + interest deductions).
4. **Less Net Upfront Bank Financing & Structuring Fees (1.00% on ₹1,875 Cr Debt)**: **-₹18.75 Crores**
5. **Less Post-Tax Processing Adjustment**: **-₹119.35 Crores**
6. **TOTAL NET STRATEGIC FINANCIAL SURPLUS**:

$$ext{NetWealthSurplus} =+ ₹375.00 ext{Cr}(Equity) + ₹225.00 ext{Cr}(ROCE) + ₹225.50 ext{Cr}(TaxShield) - ₹313.10 ext{Cr}(Adjustments) =+ ₹512.40 ext{CRORES}$$

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Module 11: Professional Certification Examination & Detailed Rationales

This module contains 10 scenario-based examination questions designed to test comprehensive mastery of Energy Sector Funding.

Question 1

Why do institutional lenders strictly size senior project debt against P90 generation probability estimates rather than P50 estimates in renewable energy funding? A) P50 represents the maximum technical capacity of the solar panels, which causes over-lending. B) P90 represents an exceedance probability of 90% (90% confidence that actual generation will meet or exceed estimates), ensuring that debt service remains fully covered even during unfavorable weather years. C) P90 yield estimates are mandated by the Ministry of Corporate Affairs for GST compliance. D) P50 yield estimates exclude transmission losses, whereas P90 includes government subsidies. * **Correct Answer**: **B** * **Detailed Rationale**: P90 reflects an exceedance probability of 90% (only a 10% risk that generation falls below this level). Sizing debt against P90 guarantees that CFADS will cover debt service even during low solar irradiation or weak wind velocity years. P50 reflects a 50% probability of default risk, which is unacceptable for senior debt.

Question 2

Under the 7-Tier Trust and Retention Account (TRA) Escrow Waterfall, which cash outflow holds absolute priority after statutory taxes? A) Senior Debt Scheduled Principal Amortization B) Essential Operation & Maintenance (O&M) Expenses, Land Lease Rentals & Insurance Premiums C) Restricted Promoter Dividend Distributions D) Debt Service Reserve Account (DSRA) Top-Up * **Correct Answer**: **B** * **Detailed Rationale**: Tier 1 covers statutory taxes. Tier 2 covers essential O&M expenses, land lease rentals, and insurance premiums. Lenders prioritize O&M and insurance over debt service because an unmaintained or uninsured energy asset cannot generate electricity, leading to total cash flow collapse.

Question 3

What is the primary credit advantage of executing a Power Purchase Agreement (PPA) with SECI or NTPC compared to a direct State DISCOM PPA? A) SECI PPAs offer 100% tax exemption under Section 80-IA. B) SECI and NTPC act as AAA-rated sovereign-backed central intermediaries with 100% Letter of Credit backup and Tripartite Payment Security Mechanisms, eliminating DISCOM default risk. C) Central utilities do not require grid connectivity approvals from the CTU. D) SECI PPAs allow the promoter to sell 50% of generated power on the energy exchange at peak tariffs. * **Correct Answer**: **B** * **Detailed Rationale**: SECI and NTPC serve as credit-worthy

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central intermediaries. Even if the end-state DISCOM delays payment, SECI/NTPC honors PPA tariff invoices to the IPP on due dates using their 100% LC backup and central payment security funds.

Question 4

What requirement must be satisfied for a commercial customer to qualify for Cross-Subsidy Surcharge (CSS) exemption under Rule 3 of the Electricity Rules, 2005 (Group Captive Structure)? A) The C&I customer must hold at least 51% equity in the SPV and consume 100% of the generated power. B) The C&I customer must hold at least 26% of equity voting shares in the SPV and consume at least 51% of the power generated by the plant annually. C) The C&I customer must be a state-owned public sector undertaking (PSU). D) The SPV must be listed on the National Stock Exchange (NSE) within 12 months of COD. * **Correct Answer**: **B** * **Detailed Rationale**: Rule 3 of Electricity Rules 2005 mandates two conditions for Group Captive qualification: (1) Captive users must hold at least 26% equity ownership in the generating plant, and (2) Captive users must consume at least 51% of total power generated annually.

Question 5

What is the mandatory timeline for filing ROC Form CHG-1 following the execution of security documents in an Energy Sector Funding facility? A) Within 180 days of the Commercial Operation Date (COD) B) Within EXACTLY 30 DAYS of security document execution under Section 77 of the Companies Act, 2013 C) Within 90 days of initial loan disbursal D) Prior to the submission of the annual GST return * **Correct Answer**: **B** * **Detailed Rationale**: Section 77 of the Companies Act, 2013 mandates that every company creating a charge on its assets must register ROC Form CHG-1 with the Registrar of Companies within 30 days of execution. Failure voids the security charge against liquidators and competing creditors.

Question 6

Under the Reserve Bank of India (Project Finance) Directions 2025/2026, what is the standard asset provisioning requirement for energy project loans during the under-construction phase? A) 0.40% B) 2.50% C) 5.00% D) 10.00% * **Correct Answer**: **C** * **Detailed Rationale**: RBI Project Finance Directions 2025/2026 mandate a 5.0% standard asset provisioning during the construction moratorium phase. Upon achieving COD, provisioning steps down to 2.5%, and further reduces to 1.0% after 2 years of operational stability (DSCR $\geq 1.25x$).

Question 7

How does Section 32 Accelerated Depreciation benefit an energy project developer under the Income Tax Act, 1961? A) It provides a 100% cash grant from the Ministry of New and Renewable Energy (MNRE). B) It allows a 40% Written Down Value (WDV) depreciation

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deduction in Year 1 on renewable energy plant and equipment, creating non-cash losses that eliminate corporate tax liabilities during initial operating years. C) It exempts the SPV from paying Goods and Services Tax (GST) on EPC contracts. D) It allows the lender to convert senior debt into equity shares at par value. * **Correct Answer**: **B** * **Detailed Rationale**: Section 32 allows a 40% WDV depreciation rate on qualified renewable energy assets in Year 1. This creates large non-cash depreciation losses, sheltering P&L operating profits from corporate income tax in early years and preserving cash within the TRA escrow.

Question 8

An energy SPV generates ₹200.00 Crores in Annual CFADS. The annual senior debt interest is ₹100.00 Crores and annual principal amortization is ₹50.00 Crores. What is the SPV's DSCR? A) 2.00x B) 1.50x C) 1.33x D) 1.00x * **Correct Answer**: **C** * **Detailed Rationale**: Debt Service = Principal (₹50 Cr) + Interest (₹100 Cr) = ₹150.00 Crores.

$$ ext{DSCR} = rac{ ext{CFADS}}{ ext{Debt Service}} = rac{₹200.00 ext{ Crores}}{₹150.00 ext{ Crores}} = \mathbf{1.3333x pprox 1.33x}$$

Question 9

What is the core purpose of a Tripartite Substitution Agreement executed in an Energy Sector Funding transaction? A) To allow the off-taker to reduce the PPA tariff if market solar prices drop. B) To grant senior lenders a 90-day cure period notice and explicit Step-in Rights to take over plant operations and substitute the promoter upon default without losing PPA tariff rights. C) To transfer 50% of project equity to the Central Transmission Utility. D) To convert floating bank interest rates into fixed-rate foreign currency ECBs. * **Correct Answer**: **B** * **Detailed Rationale**: A Tripartite Substitution Agreement binds the off-taker, lenders, and SPV. Upon promoter default, the off-taker cannot immediately terminate the PPA; it must give lenders a 90-day notice and allow lenders to exercise step-in rights to substitute the defaulting promoter with a qualified buyer while keeping the 25-year PPA active.

Question 10

Under RBI IRACP guidelines, what is the maximum DCCO extension permitted for infrastructure and energy projects due to court disputes or delayed grid evacuation without mandatory NPA classification? A) 6 Months B) 12 Months C) 2 Years (24 Months) D) 5 Years * **Correct Answer**: **C** * **Detailed Rationale**: RBI IRACP guidelines allow DCCO extensions up to 2 years (24 months) for energy and infrastructure projects facing delays beyond promoter control (such as court injunctions, land disputes, or transmission evacuation delays) without treating the account as a restructured NPA.

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